ENGR 111
Introduction to Engineering
Class #16

Presenting technical information

Good figures, charts, and tables tell a story.

Zoom protocol

Name Set your Zoom name to your real name, or whatever you'd like me to call you.

Photo Set a professional profile photo.

Camera Encouraged, not required.

Mic Off unless you're speaking.

Your questions, anytime Type in chat or raise your hand, and I'll check periodically. Or just unmute and jump in.

My questions to the class Answer in chat, or unmute and answer.

If you speak Please state your name first.

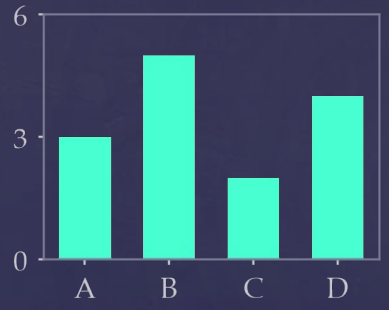
If I drop out (e.g., my Wi-Fi fails) Stay in the meeting. I'll be back in a few minutes on my phone hotspot.

Four classical chart families

There are many chart types. These four families are common in engineering.

COMPARISON

which one is bigger?



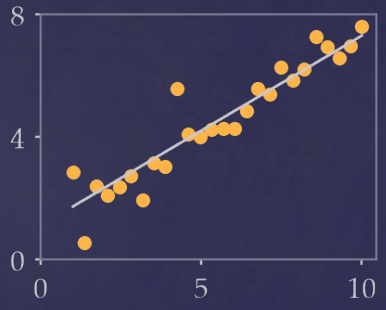
shown: bar

also grouped bar, line, dot plot

RELATIONSHIP

(and trends)

how is x related to y?

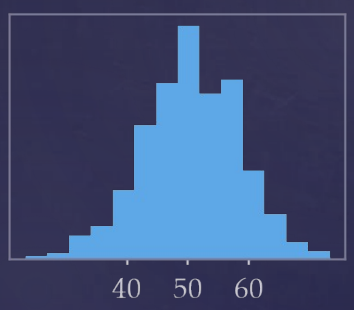


shown: scatter

also bubble, connected scatter

DISTRIBUTION

how spread out is it?

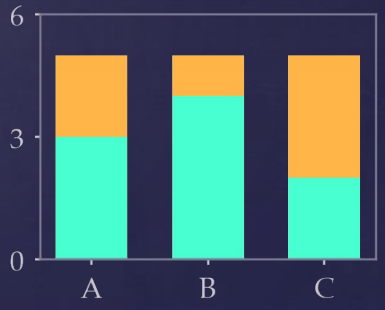


shown: histogram

also box plot, density

COMPOSITION

what are the parts of it?



shown: stacked bar

also 100% stacked, area, treemap

Start by asking what kind of comparison you are making.

Units belong in the header

A results table from a solar collector test. What does Efficiency mean?

Insolation kW	Design	Thermal kJ	Electricity kJ	Time min	Energy in kJ	Efficiency
4.8	T1	2.66	50	5	1,440	3.6569444
40.5	T1	17.1	400	7	17,010	2.452087
27.2	T1	16.2	261	8	13,056	2.1231618
12.1	T2	4.44	100	5	3,630	2.877135

What is wrong: the Efficiency column has no unit. A reader cannot tell whether 3.6569444 means 3.66% or 3.66 as a ratio.

The table gives enough to check it: 2.66 + 50 = 52.66 kJ out, from 1,440 kJ in. 52.66 / 1,440 = 0.0366 = 3.66%.

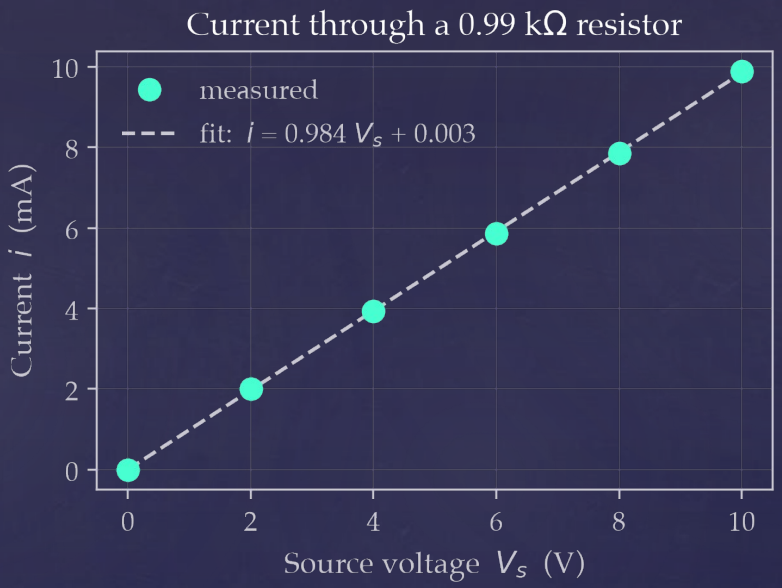
The fix: write Efficiency (%) in the header.

Then round it: 3.6569444 comes from inputs with two or three significant figures. Write 3.7%.

Solar collector test data from the course's own example set.

Label the axes, give the units

This chart works because the reader can tell what was measured.



The chart does this right: both axes are named and carry units.

The title gives the test condition: current through a 0.99 kΩ resistor.

The fitted line is printed on the chart, so a reader can use it without re-deriving it from the points.

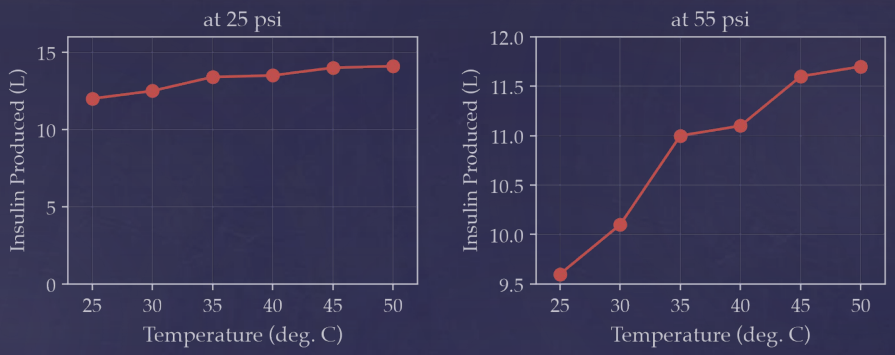
The slope is 0.984 mA per volt. What resistance does that imply, and why is it not exactly 0.99 kΩ?

Measured in class: current through a 0.99 kΩ resistor.

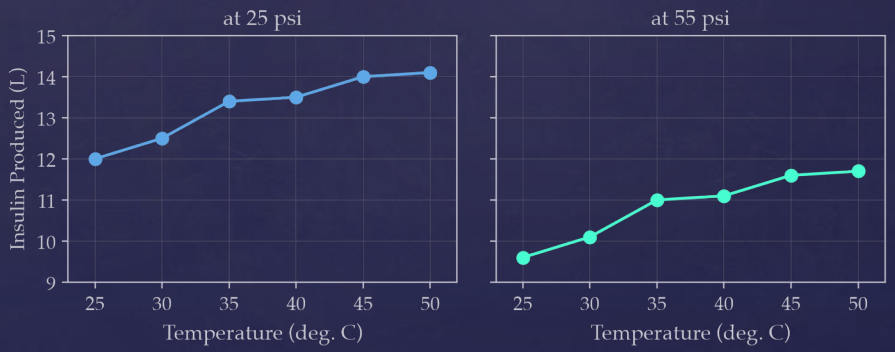
Same scale, or no comparison

Same quantity, same x-axis. Pressure is the only test condition that changes.

DIFFERENT SCALES — hard to compare



SAME SCALE — now they can be compared



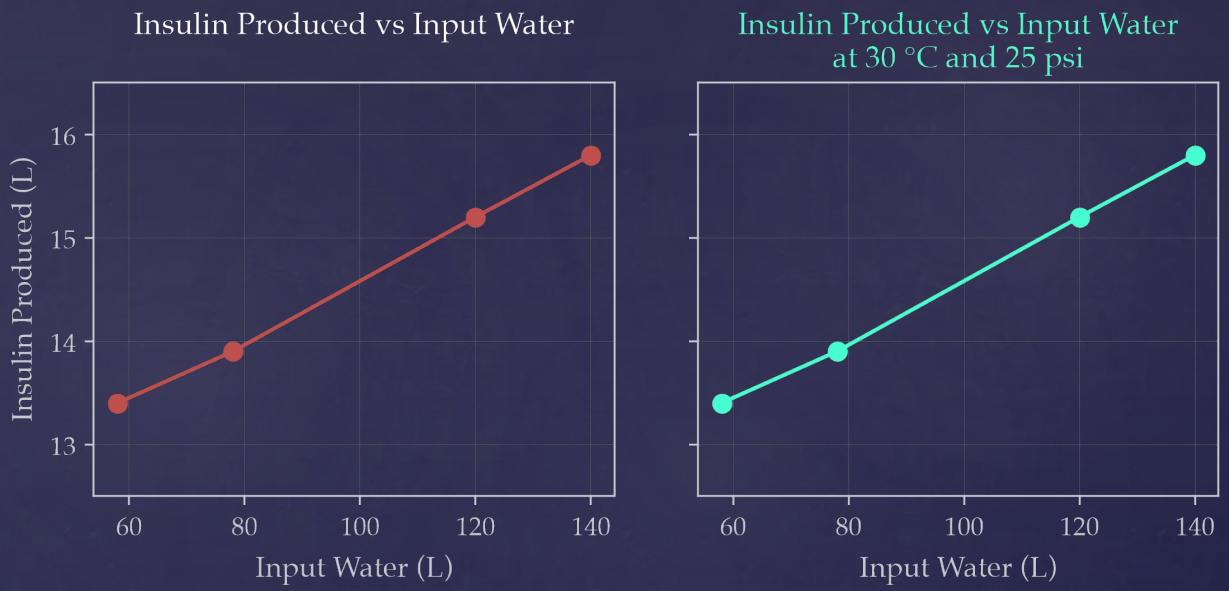
Top pair, not the same scale: the left axis runs 0-16 L, the right one 9.5-12 L. The shapes look comparable, but the levels are hidden.

Bottom pair, same y-axis: both the shape and the absolute amount are readable. At 55 psi the output is lower at every temperature.

The rule: if two plots are meant to be compared, use the same scale unless there is a clear reason not to.

Say what you held constant

The same chart twice, with and without the test conditions.



The left chart is wrong: water was varied while temperature and pressure were held still, and nothing on the chart says so.

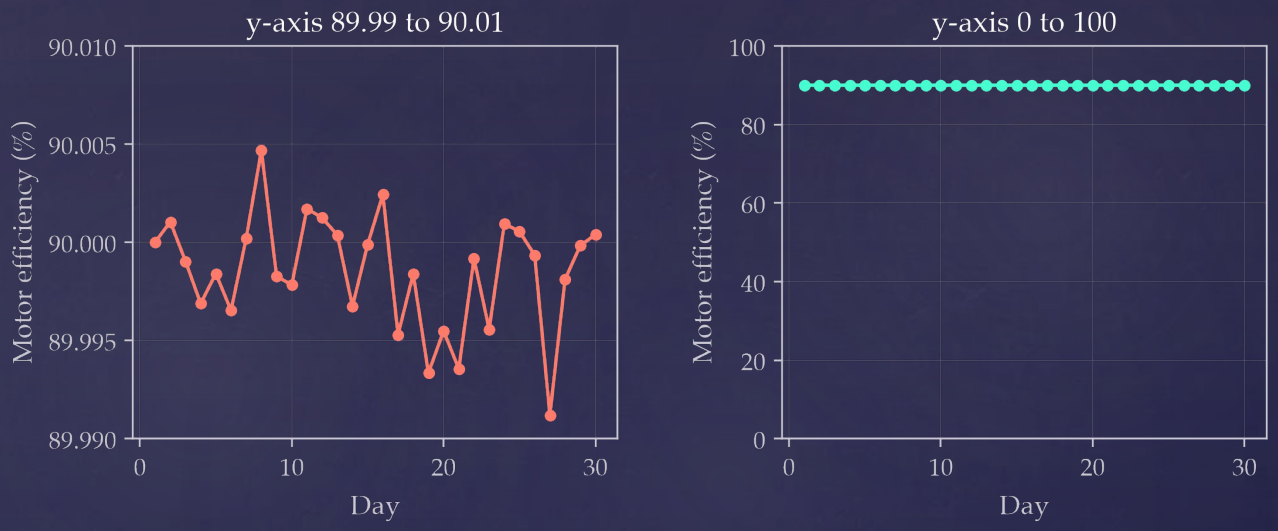
A reader cannot tell whether the other two moved, so the result cannot be trusted or repeated.

The right chart is the fix: the held values go in the title, at 30 °C and 25 psi.

The held values do not have to be printed on the plot. A figure caption works just as well. They just have to be somewhere a reader will find them.

Scale decides what noise looks like

One motor, one month of efficiency readings, plotted twice.



Same file, both charts. The left one looks like a failing machine. The right one looks perfectly steady.

Both plots are the same data. The y-axis range decided what the reader notices.

Pick the range from the tolerance. If the spec is 90 ± 0.005%, the left chart is the correct one, and it should have the limit lines drawn on it.

Bar charts are different. A bar's length represents the value, so cutting the axis can mislead.

Nine questions to ask of a figure

Eric's checklist. From here on, one real graphic at a time.

Does it tell a story? What is the point? Decide that before you draw anything.

Is it informative? It should say more than a sentence would.

Is it concise? Combine charts that belong together. Cut what is not used.

Is it readable? Labels, units, context, and large enough text.

Is it simple and uncluttered? No chart junk, no 3-D, no decoration.

Is it aesthetically pleasing? Color, contrast, composition, coherence.

Does it target the audience? Engineers, or the public, or a manager.

Does it have an impact? Is the takeaway clear?

Does it complement your discussion? A figure needs your words with it.

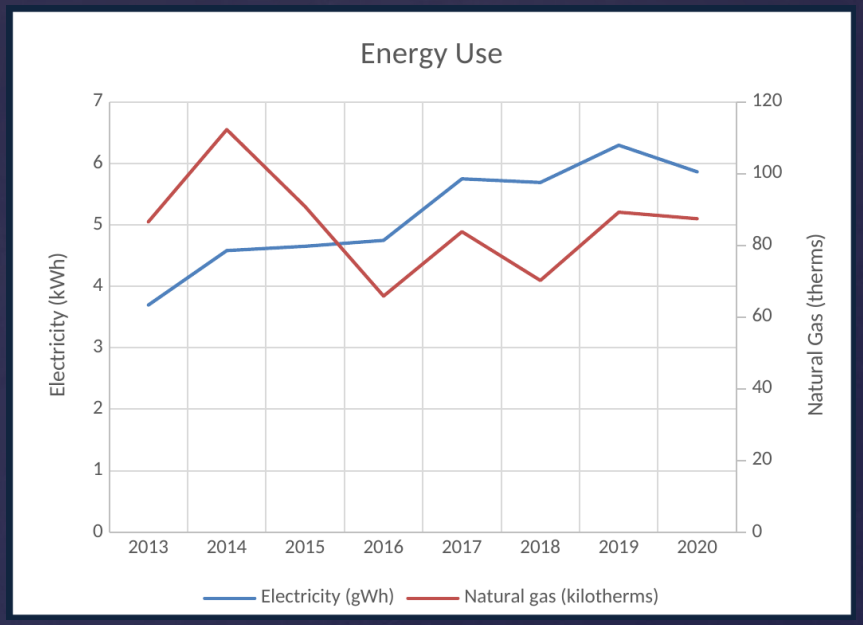
Most figures fail more than one question. Start with the most obvious one.

These nine are Eric Lee's, from the course visualization handout. His list, not a standard one.

Does it tell a story?

Question one. Table or chart? The same eight years, shown both ways.

Year	Electricity kWh	Natural gas therms
2013	3,695,610	86,571
2014	4,581,506	112,290
2015	4,652,583	90,807
2016	4,748,628	65,864
2017	5,749,628	83,818
2018	5,690,549	70,216
2019	6,296,512	89,253
2020	5,863,727	87,436



Neither form is wrong. They answer different questions.

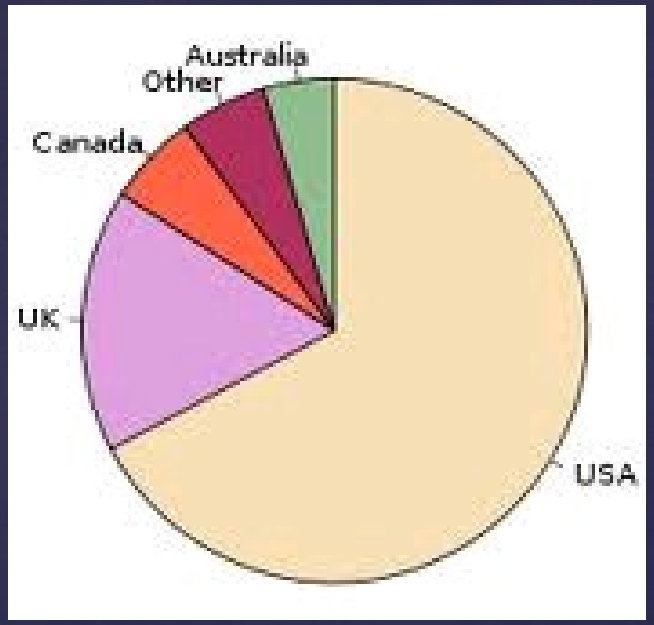
A table, if a reader has to look up what 2017 actually was.

A chart, if the point is whether use went up.

Which one would you put in your report, and what would you have to say out loud to go with it?

Is it informative?

This one fails in many respects.



No title, no units, no source. Nobody can say what is being counted.

No slice carries a number, so you cannot read a value off it. All you learn is that the USA is more than half of something.

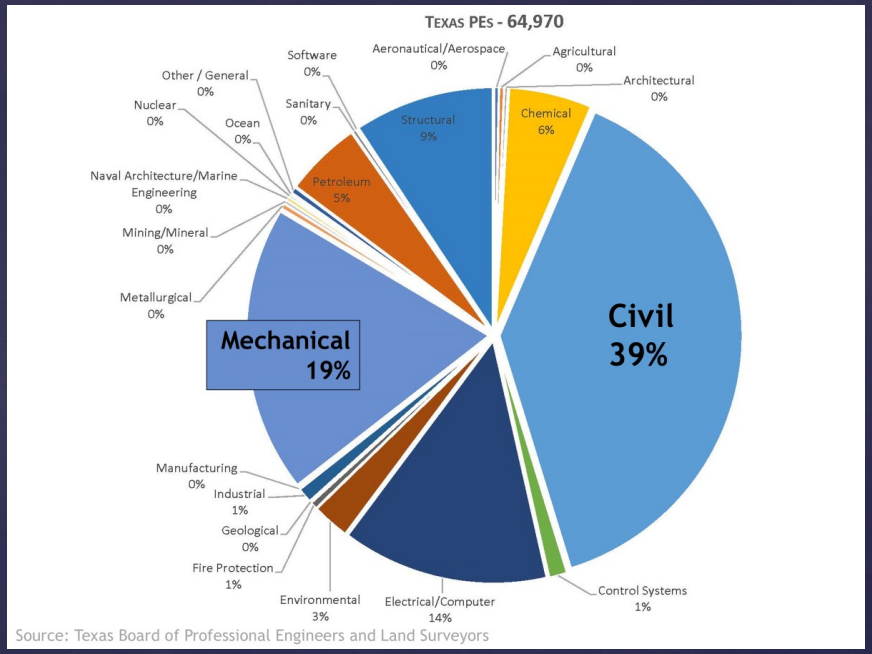
The slice count is fine. Five slices is within the rule, so that is not the problem here.

The fix: give it a title and source, and put the percentages on the slices or in the caption.

Wikipedia, "Pie chart."

Is it informative?

Also fails, and this one does say what it is.



What it gets right: a title, a total, a source line, and percentages where a reader can find them.

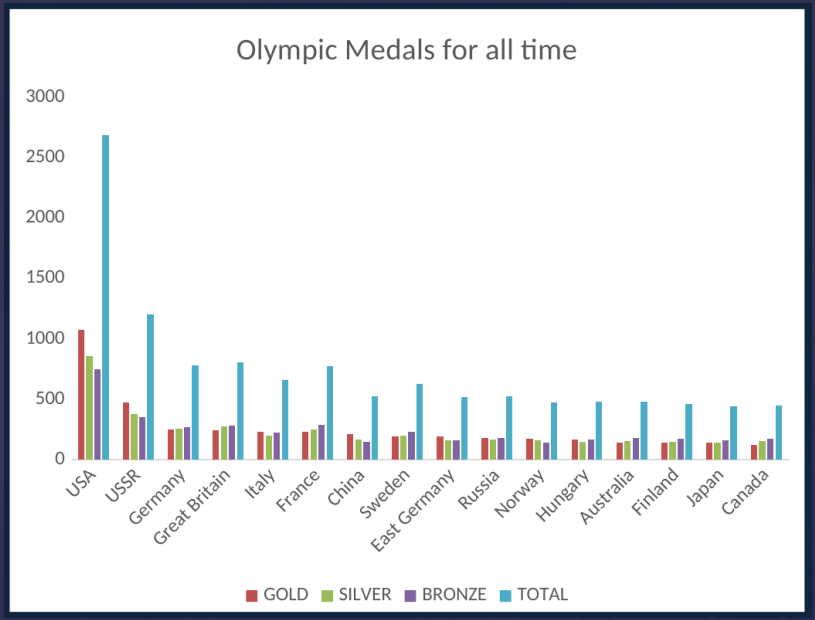
What is wrong: thirteen slices read 0% and each one still spends a label and a leader line, which buries the content.

The fix: a bar chart with everything under 2% collected into "other". Then civil at 39% reads immediately.

Texas Board of Professional Engineers and Land Surveyors.

Is it concise?

This message could be conveyed better with stacked bars.



What is wrong: the fourth bar, TOTAL, is just the other three added up, so it carries no new information.

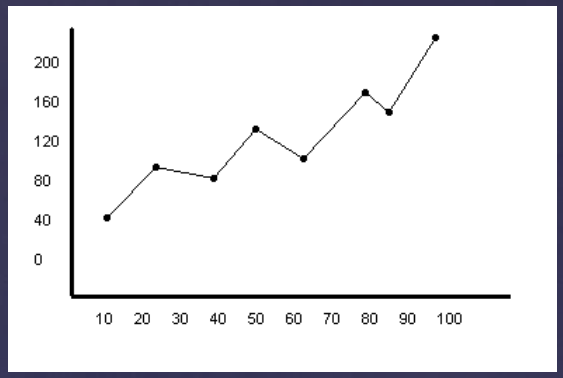
And it does damage. At more than twice the height of the tallest real bar it squashes gold, silver and bronze into the bottom of the chart.

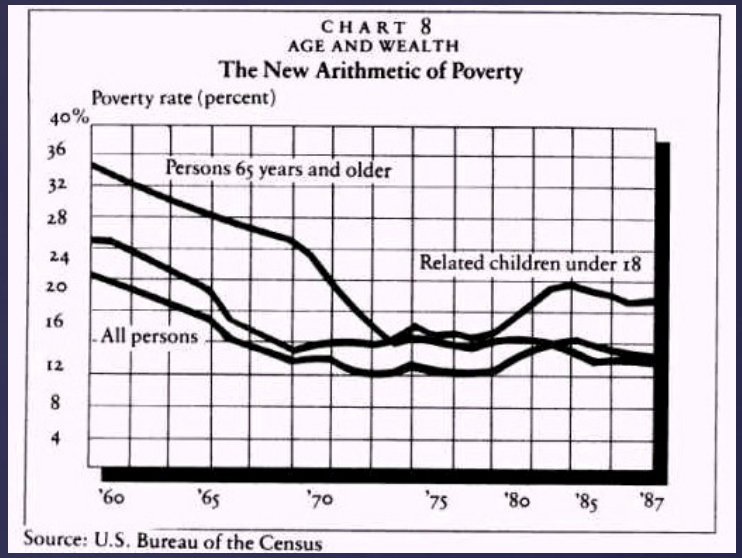
The fix: one stacked bar per country. The total is the bar's full height and the split is visible inside it.

All-time Olympic medal counts, olympic.it.

Is it readable?

Both of these fail, and they fail differently.



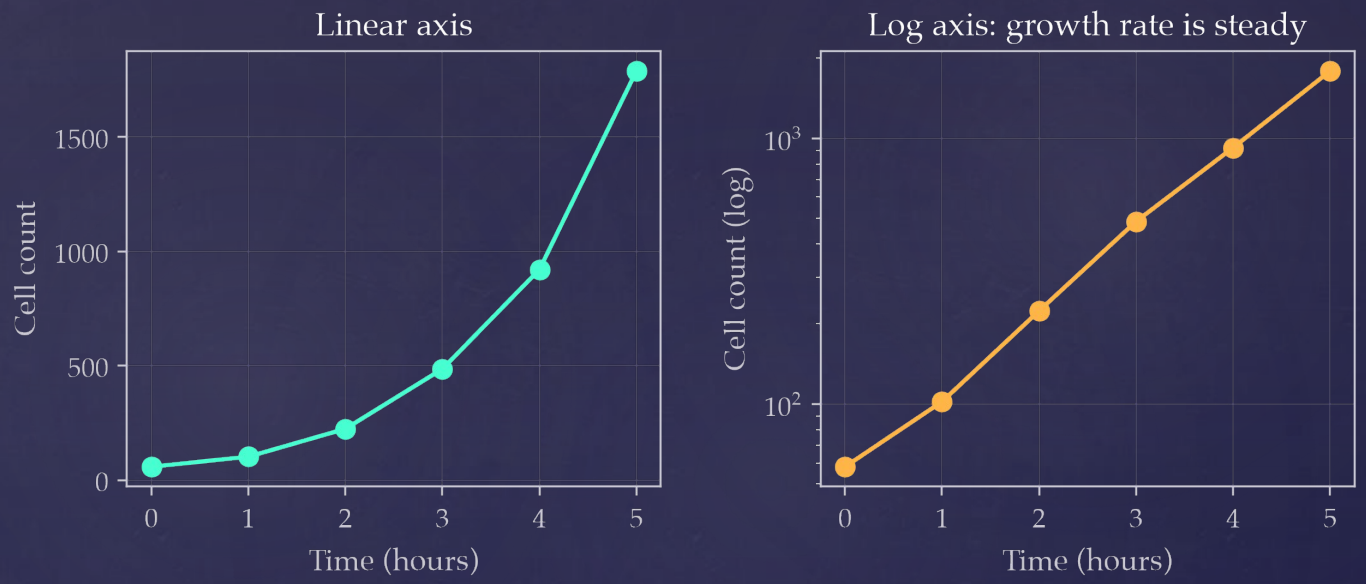


Left: no labels. Nothing says what either axis measures, or in what units.
Left: the x-axis never reaches zero, and nothing says so.
Left: joining the points claims readings between them that nobody took.
Right: you cannot tell the lines apart, and the heavy grid and shading fight the data.
Right: the spacing is erratic. '60 to '65 is five years. '85 to '87 is two.
Which of the two is worse, and could either one be rescued without collecting new data?

Bad-chart examples collected by G. M. Klass, Illinois State University; right-hand chart, U.S. Bureau of the Census.

Is it readable?

This one works. Cells in an agar culture, hourly.



What both charts get right: each axis is named and carries its unit, so a stranger can read them.

The left chart still hides the finding. On a linear axis the early hours are pressed flat against the baseline.

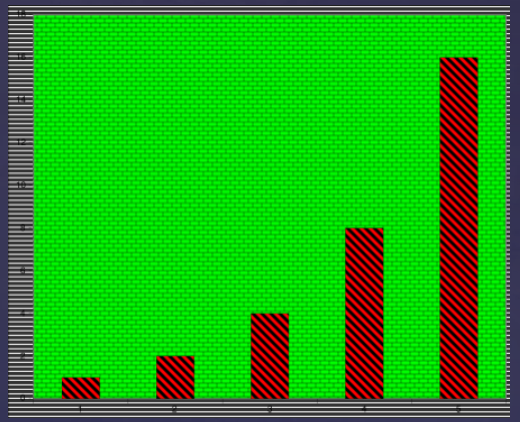
The right chart is better: on a log axis the points fall on a straight line, which is what a constant growth rate looks like.

Same numbers both times. The axis choice decided what you could see.

Agar culture cell counts, from the course's own in-class data set.

Is it simple? Uncluttered?

All three fail. Texture and 3-D graphics detract from the information content.



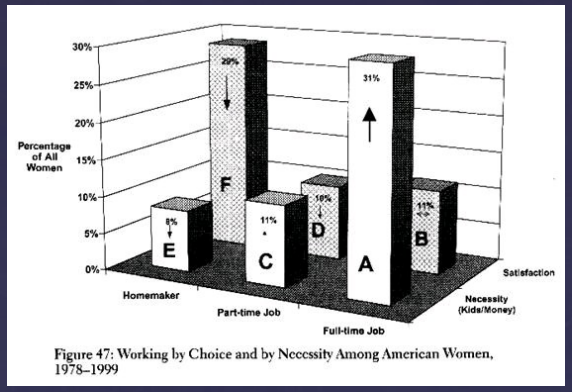
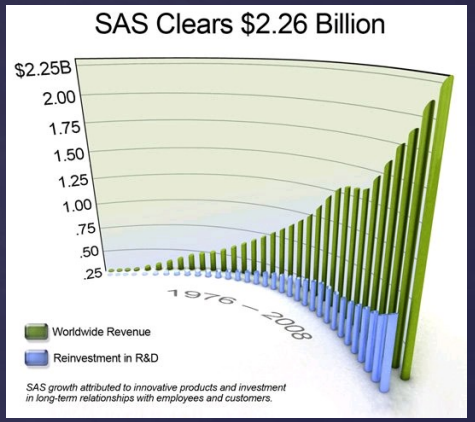


Figure 47: Working by Choice and by Necessity Among American Women, 1978–1999



Left: five numbers. The green field, the hatched border and the stripes carry no data at all. Delete every one of them.

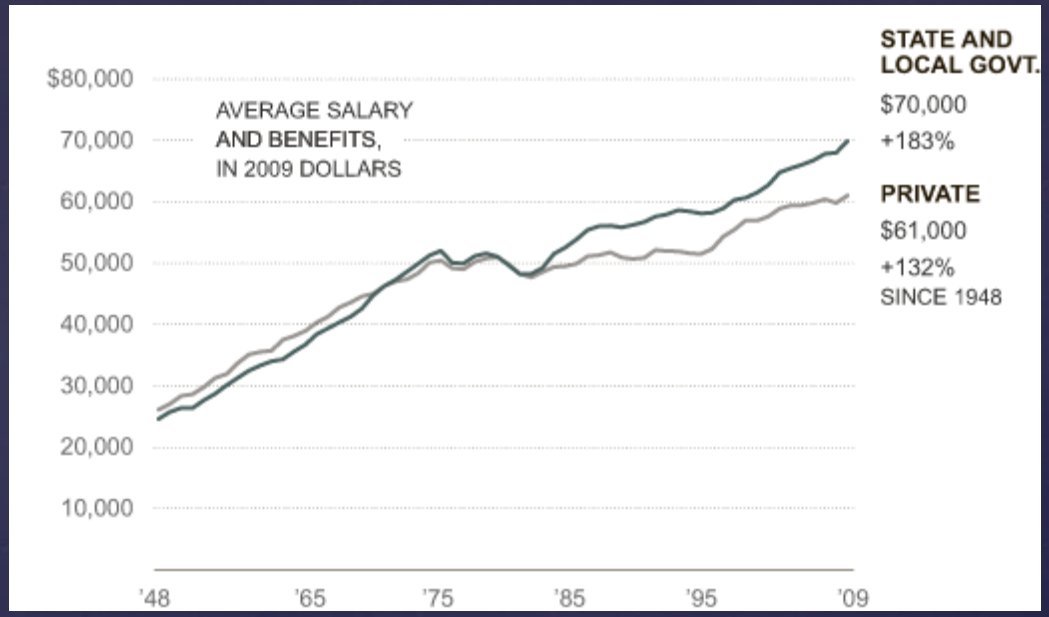
Middle: 3-D bars. Try to read a value off one - the front face and the top face give different answers. Use flat bars.

Right: a curved ribbon, so the same quantity looks bigger or smaller depending where it sits along the curve. Straighten the axis.

Wikipedia, "Chartjunk"; G. M. Klass; and a corporate annual report.

Is it simple? Uncluttered?

This one mostly works. Same job as the last slide, done carefully.



What it gets right: the two series are labeled at the ends of the lines, where your eye already is, so there is no legend to look back at.

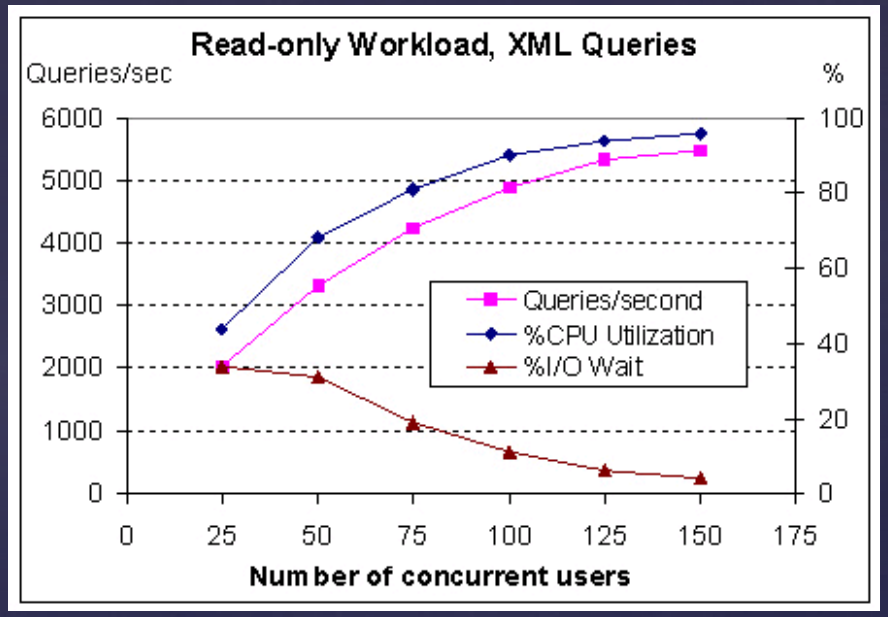
The grid is faint and dotted, so it stays behind the data instead of competing with it.

The block on the right gives the 2009 value and the growth since 1948, so the reader does not have to measure anything.

Its one weakness: those two labels line up with the two lines only by height. Putting each label right at its own line end would fix it.

Is it aesthetically pleasing?

Close to the old Excel default. This one is worth arguing about.



It is easy to call this ugly. Magenta on maroon, dashed grid, a legend parked in the plot.

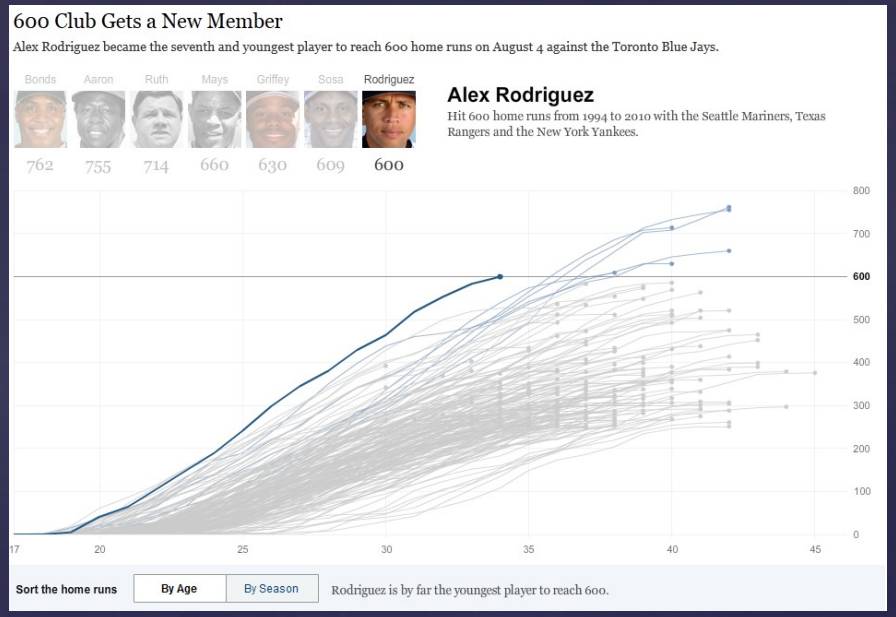
But it is clear. Axes named with units, and you can read a value off it.

The markers are different shapes, so it still works printed in black and white. That is a real strength.

Aesthetics is the softest of the nine. So is this a bad chart, or just an ugly one?

Is it aesthetically pleasing?

This one is interactive, and the graph is clean and easy to look at.



What it gets right: hundreds of gray lines and one blue one, so you know instantly which player is the subject.

The gray lines give context: 600 home runs means more when every other player's line is visible behind it.

The technique to copy: push everything that is background into gray, and give color to the one series you are talking about.

nytimes.com.

Is it aesthetically pleasing?

A beautiful figure still has to be readable.



Every choice here is deliberate, and the result is genuinely beautiful.

Now try to use it. Pick one cause and follow its line to a feeling. You cannot, because every line crosses every other line.

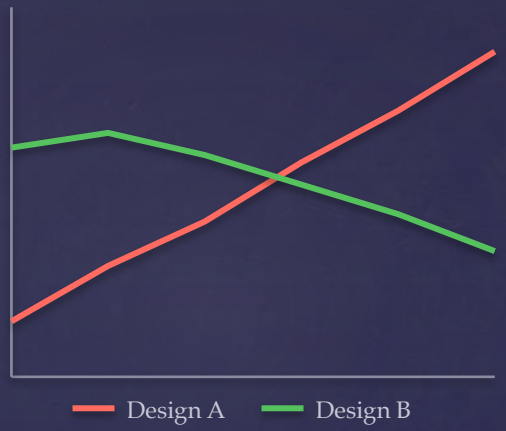
Where it went wrong: the design became the point. Pretty and unreadable is still a failed figure.

brainpickings.org.

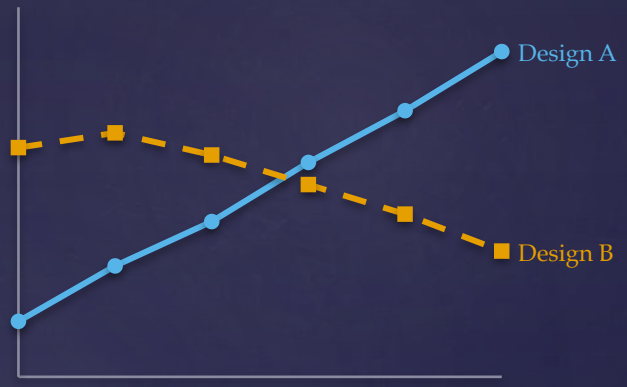
Don't let color do the work alone

The standard is blunt about it: color is never the only thing carrying meaning.

Color only



Color plus a second channel



About 8% of men have red-green color vision deficiency, and about 0.5% of women.
Add a second channel. A dashed line, a different marker shape, a pattern in the bar.
Label the line at its end instead of using a legend. Then the color is decoration.
Keep the contrast up. Pale yellow on white fails for everybody, not just some readers.
Use a tested palette. Okabe-Ito was built for this: 0072B2, D55E00, 009E73, E69F00.
Print it in gray. If it still reads, you are done.

nei.nih.gov, WCAG 2.1, Okabe and Ito.

Does it target the audience?

Readable if you work in finance. Otherwise it is a wall of lines.



For its intended reader this works. Candlesticks, moving averages and a volume panel, all axes labeled.

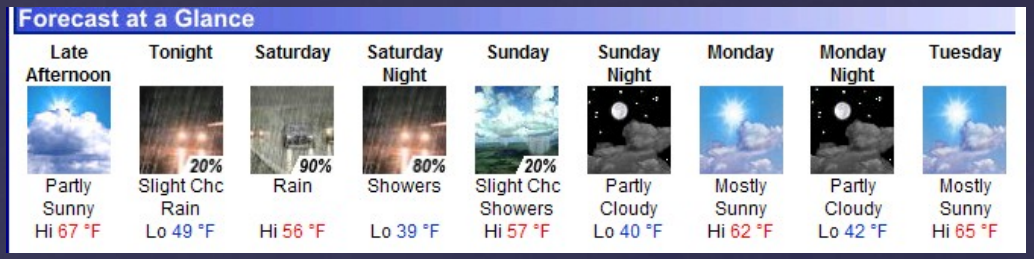
For anyone else it fails. Nothing on it explains what a candlestick is, so the detail is wasted.

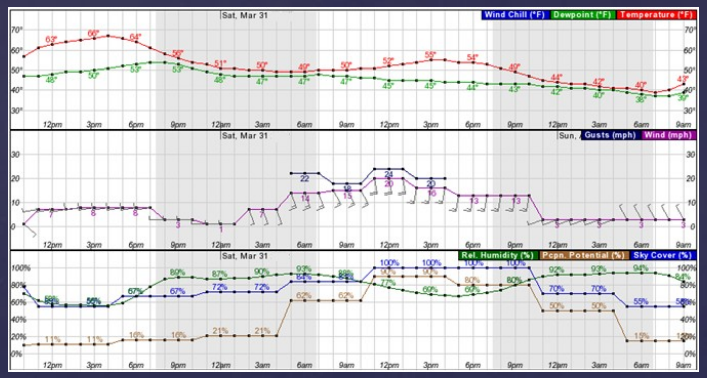
The point of showing you this: a lot of engineering plots look exactly like this to people outside the field. Know who is reading.

stockcharts.com.

Does it target the audience?

The same forecast, published twice by the same agency.





The top strip is for everybody: a picture of a cloud, a high and a low, and nothing else.

The bottom one is for a specialist: dewpoint, wind barbs, sky cover, hour by hour.

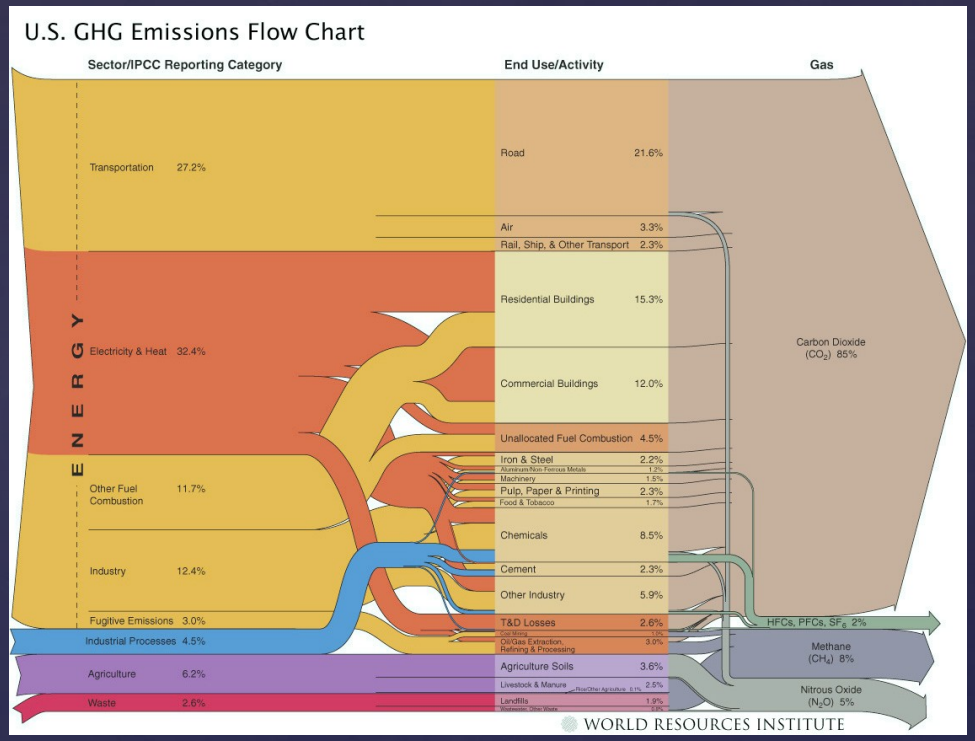
What they got right: they did not compromise between the two readers. They published a version for each.

Which one belongs in a live talk, and which belongs in a written report?

noaa.gov.

Does it have an impact?

Clear takeaway information, and it is memorable.



This is a Sankey diagram. Band width is quantity, and every flow has to balance, so nothing can go missing.

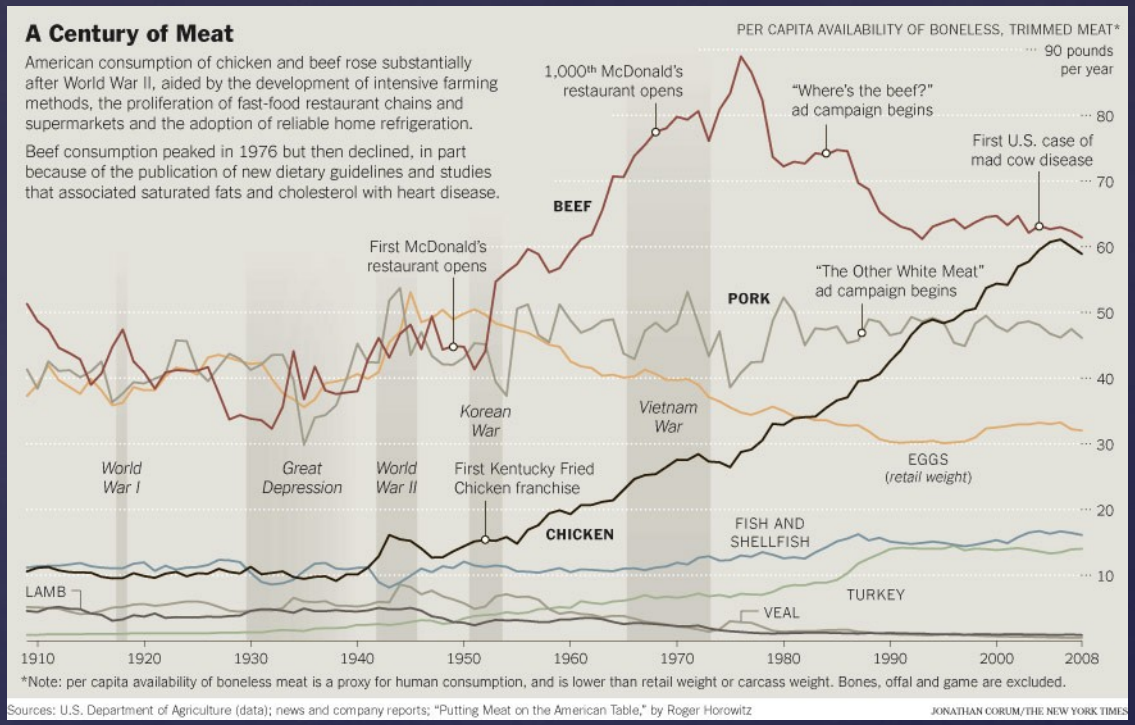
What it gets right: you can follow one band the whole way across - transportation, into road, into carbon dioxide - and read its share.

Why it is here: it shows three linked variables at once, which bars would struggle to do

U.S. greenhouse gas flows as a Sankey diagram. World Resources Institute.

Does it complement your discussion?

A blend of narration and data visualization can be very powerful.



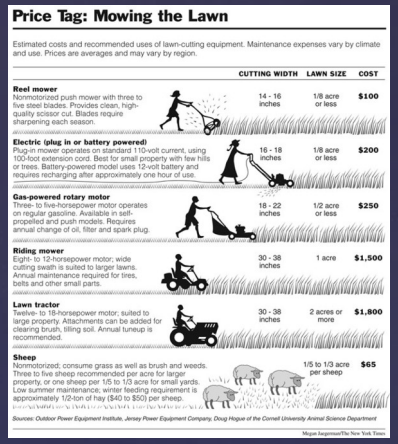
What makes this work: the important turns in the line are labeled right where they happen.

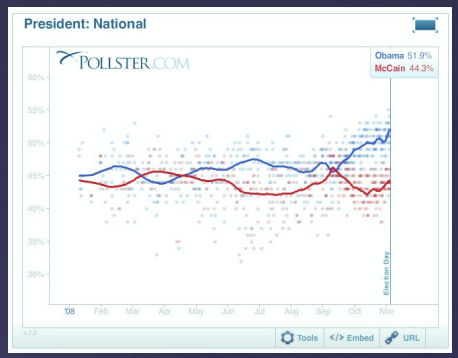
Each turn in the data gets a cause: the first McDonald's, mad cow disease, the wars shaded in gray.

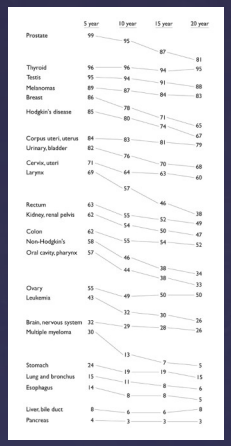
What to copy: when a line changes direction, say why on the chart. A figure with no words around it makes the reader guess.

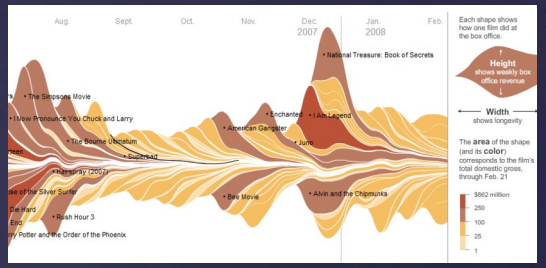
Jonathan Corum, nytimes.com.

Be creative!









Category	Age						Education				Gender		Children		Browse Location		
	18-24	25-34	35-44	45-54	55-64	65+	No College	Some College	College	Grad School	Male	Female	Children	No Children	Home	School	Work
Gaming	30	5	-10	-18	-30	-19	6	8	-0	-4	21	-12	-8	13	14	0	-7
Creative Arts	19	8	-6	-11	-15	-14	-5	7	6	-7	6	-1	-12	17	6	1	-4
Info-sharing	11	6	-6	-6	-9	-14	-8	4	-0	1	-1	3	-9	12	1	-12	3
Education	9	6	-2	-7	-10	-10	-0	7	6	-6	12	-4	-15	22	9	-3	-3
Places	-4	17	-2	-7	-9	-12	-17	-11	2	12	-6	11	-17	25	-10	-15	16
Lifestyle	3	10	-1	-4	-9	-10	-12	-0	8	-0	-17	27	-9	15	-1	-9	6
Friends	8	4	-2	-3	-10	-7	-4	7	3	-2	5	1	-9	14	3	-6	2
Interests	4	6	-4	-3	-6	-8	-6	1	7	-3	11	-2	-8	12	-2	-7	6
Dating	12	-0	-3	-1	1	-0	-14	0	1	5	-11	20	-11	15	5	3	0
Business	-8	7	1	5	-1	-7	-17	-3	4	6	-0	5	-6	8	-10	-11	18
Family	-11	4	6	5	4	1	-3	4	7	-6	-20	31	-11	-2	8	-10	-3
Total	7	7	-3	-4	-8	-10	-8	2	4	-9	-0	7	-9	14	1	-6	4

A table designed for comparison, not just exported from Word.

A slopegraph: two columns of numbers, joined so you can see which rose.

The caveat: beware gimmicky graphics that are hard to understand.

edwardtufte.com, nytimes.com, olinmontalvo.com.

Example

The same data set four times, each version a step better than the last.

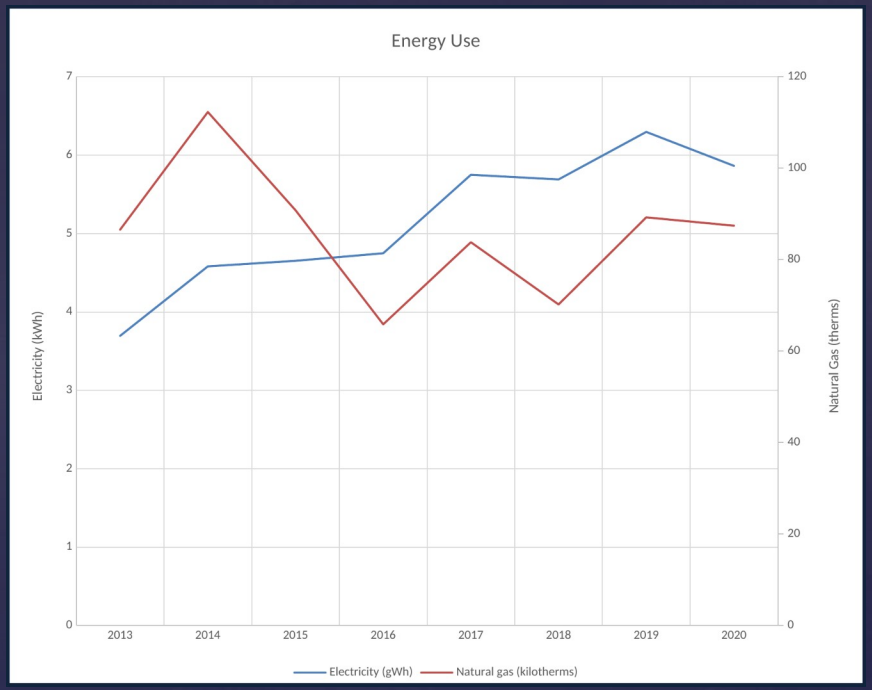
Year	Electricity kWh	Natural gas therms
2013	3,695,610	86,571
2014	4,581,506	112,290
2015	4,652,583	90,807
2016	4,748,628	65,864
2017	5,749,628	83,818
2018	5,690,549	70,216
2019	6,296,512	89,253
2020	5,863,727	87,436

If the numbers are not the message but the trends are, this belongs in a figure.

So start by asking what the message is. Not "here is our energy use." The message is that something changed.

Example

Select the table, press the chart button. This is what you get.



What it gets right: the table became a chart, and both fuels are visible.

What is wrong: two units are plotted on two different y-axes.

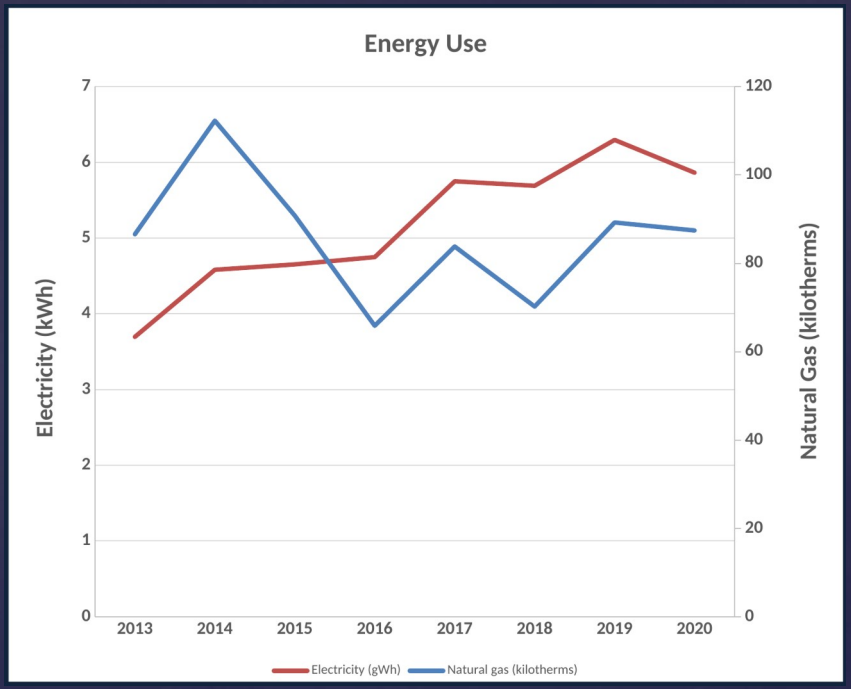
Where the lines cross means nothing. That crossing comes from where Excel put the two scales.

You keep going back to the legend to remember which line is which.

The units are unclear. The table is kWh and therms, but the chart values have been scaled down.

Example

Somebody spent five minutes on it.

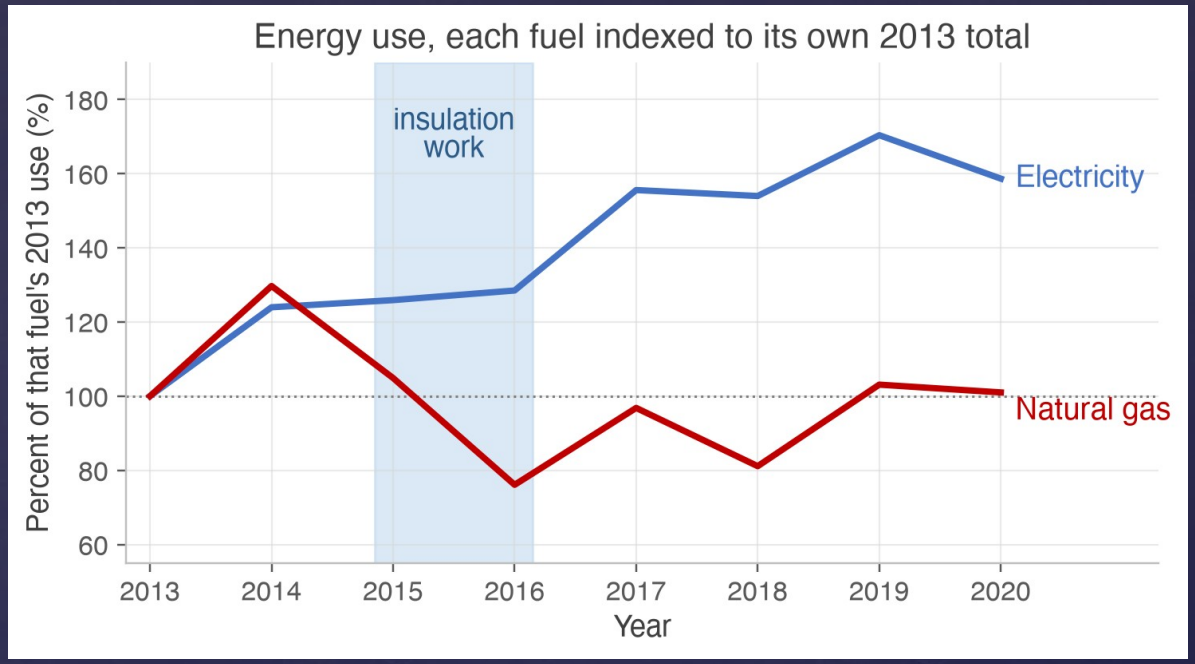


Bigger type, most of the grid gone, both axes named. Genuinely better, and most work stops before here.

But the real problem has not moved. Still two scales, so the two lines still cannot be compared.

Example

Change what you plot, not how it looks.



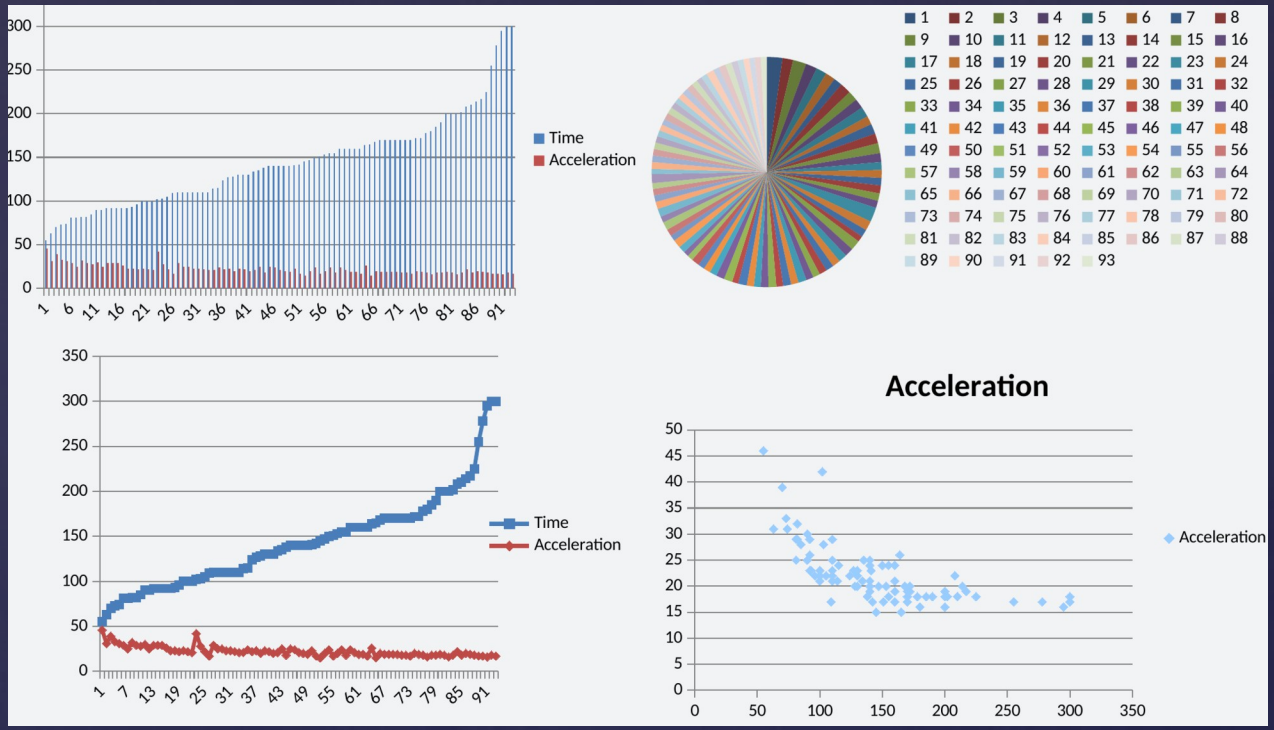
Each fuel is divided by its own 2013 value. This is an index, not a share of total energy. The two lines do not add to 100%.

What the plot shows: gas fell during the insulation period, but electricity kept rising after it.

What is still not great: It doesn't give a sense of why electricity went up when natural gas went down. Did it have to do with the insulation work? This can be explained in the text accompanying this plot.

This is not easy, though

Four graphs of the same data, each made quickly with Excel.



Find the pie chart. Ninety-three slices, and a legend taller than the plot.
Nothing here threw an error. The software will not stop you.
A good figure takes a second pass, every time.

Good figures, charts, and tables

The same nine, and they are in the handout.

Tell a story.

Are informative.

Are concise.

Are readable.

Are simple and uncluttered.

Are aesthetically pleasing.

Target the audience.

Have an impact.

Complement thoughtful discussion.

Which one would you check first on a figure you made quickly?